PROJECT BRIEF

Hypermarket Camera Analytics & Smart Operations

A practical plan for customer-flow insights, produce availability, POS-linked sales analysis, and human-reviewed loss-prevention alerts.

Purpose

Turn IP-camera video and store data into useful operational signals for managers and staff.

Recommended starting point

Pilot one produce area, one fixed camera, a few mapped bins, and a POS data feed. Keep security review as a separate, controlled feature.

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Camera models, POS vendor, country, and store scale remain to be confirmed.

1. Executive summary

The web app combines fixed IP-camera analytics with POS, inventory, and staff-confirmation data. It helps store teams understand where and when customers spend time, estimate visible produce availability, compare those conditions with sales, and review potential security events.

Cameras provide estimates and review cues. POS and inventory remain the source of truth for transactions and on-hand stock; authorized people make security decisions.

Business outcomes

- Reduce time produce displays remain visibly low or empty.
- Understand section traffic, dwell patterns, and peak staffing periods.
- Compare traffic, display availability, sales, and promotions by product/time.
- Help loss-prevention staff find relevant footage faster, with human review.
- Improve estimates using verified examples rather than unreviewed model predictions.

Scope boundaries

- No face recognition, persistent shopper identity, or automated accusation.
- Camera cannot see back-room stock or reliably establish an individual purchase.
- Ordinary video cannot promise exact weight or count for loose produce; use scales and inventory records.
- Local legal review, notices, access policies, and retention rules are required before launch.

Recommended first release

Capability	Pilot deliverable
Customer flow	Anonymous zone visits, hourly counts, average/median/maximum dwell estimates.
Produce availability	Per-bin visible fill estimate and OK/low/likely empty/view blocked state.
Sales context	POS units and sales by time bucket and mapped product/category.
Security review	Staff-reviewed event clips after separate controlled evaluation.

2. Camera opportunities and limits

A camera may support several analytics only when its position and image quality suit each task. Define each camera purpose and image region explicitly. Do not assume every camera is suitable for every feature.

Use case	Camera output	Data / limit
Produce traffic	Zone entries, visits per period, dwell distribution.	Occlusion and track breaks affect accuracy.
Produce availability	Visible coverage/fill per mapped bin; possible low/empty state.	Hidden produce and back-room inventory are not visible.
Restock operations	Likely transition from low to replenished.	Staff confirmation or inventory events verify it.
Sales/promotion	Sales and units compared with traffic/availability.	POS and promotion data; correlation is not individual attribution.
Checkout queues	Queue size and estimated wait.	Needs a dedicated, clear view and local validation.
Safety/housekeeping	Possible spill, blocked path, fallen produce cue.	Human verification required.
Loss-prevention review	Possible item concealment sequence for staff review.	High false-positive/miss risk; not proof of theft.

Camera setup

- For bins, prefer a stable elevated view that sees the bin interior; map one region per bin.
- Use dedicated views for queues or security rather than stretching a produce camera across unrelated areas.
- Record camera ID, field of view, region polygons, time sync, stream health, and purpose.
- Check ONVIF/RTSP compatibility and supported codecs. ONVIF Profile T covers advanced streaming including H.264/H.265 and RTSP capabilities.

Possible later modules

After the pilot is reliable, consider replenishment prioritization, queue alerts, display compliance, blocked-aisle prompts, spill cues, and demand forecasting. Add each only after defining its owner, validation method, and data purpose.

3. Functional requirements and roles

ID	Requirement	Expected behavior
FR-01	Camera health	Show online/offline, last frame, stream errors, and configured regions.
FR-02	Traffic	Aggregate counts and dwell summaries by store/zone/date/time bucket.
FR-03	Bin mapping	Admin maps image region to product/bin and threshold.
FR-04	Availability	Store fill estimate, confidence, status, time, and model version.
FR-05	Alerts	Require persistent condition; deduplicate; allow acknowledge/resolve.
FR-06	POS import	Import timestamped product sales, units, amounts, and store identifiers.
FR-07	Inventory	Optionally ingest receiving, on-hand, waste, and restock records.
FR-08	Feedback	Staff confirms/corrects status with user and timestamp.
FR-09	Security review	Restrict event clips by role; log view/export/action.
FR-10	Audit/export	Export aggregate reports; version configuration and models.

User roles

Role	Typical access
Store manager	Dashboard, aggregate reports, stock alerts, task status.
Department associate	Assigned restock/check tasks and status confirmation.
Loss-prevention reviewer	Restricted alert queue and relevant clips; review outcome.
System administrator	Camera setup, roles, integrations, thresholds, retention.
Analyst	Aggregated reports; no security-clip access by default.

4. Architecture and data flow

Keep video processing close to the cameras where practical. Send structured events to the application, and separate ordinary analytics from any restricted security footage. Deployment may be local, cloud, or hybrid depending on network, latency, and policy.

Layer	Responsibility	Example
Camera	Authorized video stream and health.	IP camera; RTSP/ONVIF; segmented network.
Video processing	Sample frames; detect/track people; estimate bin status.	Edge PC/GPU or local video server.
Integration	Normalize POS/product/inventory/restock feeds.	API, secure file transfer, scheduled import.
Backend	Auth, APIs, alert rules, audit, retention.	Web service, job queue, role-based access.
Data store	Aggregate time series and event metadata.	Relational/time-series store; restricted object storage.
Web app	Dashboards, maps, alerts, tasks, exports.	Responsive browser UI and charts.

Flow

IP camera -> local processor -> anonymous aggregate events and bin estimates -> application database -> dashboard/alerts. POS and inventory data enter separately and are joined at store/product/time-bucket level. Restricted security events use a separate access-controlled review process.

Processing pattern

- Analyze selected frames/windows when real-time video is unnecessary.
- Synchronize camera processor, POS, and inventory timestamps.
- Keep raw video separate from aggregate analytics; restrict and audit any clip access.
- Store model version, camera configuration, threshold, and confidence with events.

5. Model design and learning loop

Use separate analytics tasks. A person tracker does not solve produce recognition, fill estimation, or security-event review.

Task	Approach	Output
People and dwell	Person detector, temporary multi-object tracker, entry line and zone.	Visit, dwell estimate, confidence, time bucket.
Produce display	Mapped-region classifier/segmenter trained on store images.	Product/bin, visible fill, state, confidence.
Restock change	Compare stable bin states over time; combine with stock events.	Possible restock time and confidence.
Security review	Research-stage short-clip action cue; human review.	Restricted review event, never a theft verdict.
Forecasting	Time series using verified sales, availability, traffic, calendar.	Demand/restock priority estimate with uncertainty.

Training and evaluation

- Collect samples across busy/quiet periods, lighting, camera conditions, and display layouts.
- Label representative frames/clips for people counts, bin states, and review outcomes using clear definitions.
- Split examples by date/camera/store to test generalization.
- Track count error, fill error, precision/recall, false alerts, missed events, and confidence calibration.
- Use reviewed errors to improve camera placement, region mapping, thresholds, and model.
- Version labels/data, model weights, thresholds, and deployment; roll back regressions.

Controlled learning

Do not automatically treat model predictions as correct labels. Use staff-confirmed cases and sampled audits as verified labels, retrain offline, evaluate on held-out examples, then deploy a versioned model.

6. Loss-prevention review module

The model may flag an ambiguous sequence, such as an item being picked up and moved toward a bag or pocket, followed by the person leaving the display area. Shoppers also perform ordinary actions that can look similar. The output must be described as a possible concealment cue for review, never a theft determination.

Human decision required: no automatic accusation, stop, search, denial of service, or law-enforcement referral from a model alert alone.

Review workflow

Possible event -> restricted alert with camera/time/reason -> authorized reviewer checks context and relevant clip -> reviewer records no concern, needs follow-up, or confirmed incident under store policy -> retention follows approved schedule. Confidence is not proof.

Limitations

- Hands, bags, carts, crowds, and occlusion make interactions hard to classify.
- Video may not show whether an item was later paid for, returned, or transferred. POS context is not proof of intent.
- Wide views cover more area but may reduce detail; placement must match the purpose.
- Monitor false alerts and missed events; stop or redesign if the workflow is not useful.
- Do not use face recognition or persistent biometric identity.

Pilot gate

Run a shadow evaluation first: create alerts without notifying floor staff, then have authorized reviewers record false positives, misses, and usefulness. Only consider a limited live review queue after purpose, staffing, privacy review, policy, and acceptable alert quality are approved.

7. Data model and dashboard

Use store/product/time aggregates. Do not link an individual temporary camera track to a receipt or long-term shopper profile.

Entity	Key fields
Camera	camera_id, store_id, stream_ref, location, status, purpose, retention policy
Zone/bin	zone_id, camera_id, polygon, product_id, capacity reference, thresholds
Traffic aggregate	store_id, zone_id, bucket_start, visits, avg/median/max dwell seconds
Availability	store_id, zone_id, product_id, observed_at, fill_estimate, state, confidence, model_version
POS aggregate	store_id, product_id, bucket_start, units_sold, sales_amount, promotion_id
Inventory event	store_id, product_id, event_time, event_type, quantity, source
Alert/review	alert_id, type, created_at, camera/zone, confidence, reviewer, outcome, audit ref

Example aggregate

```
STORE_01 | 2026-10-07 10:00 | FRUIT_A | APPLE_RED | visits=42 | avg_dwell=68s |
max_dwell=241s | fill=0.32 | LOW | confidence=0.87 | units_sold=18
```

Fill is a visible coverage estimate, not scale weight or exact quantity.

Dashboard screens

Screen	Content
Store overview	Current alerts, cameras online, sales, traffic, department status.
Produce map	Bin/product, fill estimate, status, confidence, last checked, restock action.
Customer activity	Traffic by hour/day, dwell distribution, peak periods, zone comparison.
Sales and availability	Product sales, units, traffic, promotions, low-stock timeline.
Staff tasks	Restock/check queue, priority, age, acknowledgement, completion.
Security review	Restricted queue, event time/camera/reason, controlled clip, reviewer outcome.
Administration	Camera mapping, roles, thresholds, integration status, retention.

8. Alerts, integrations, security, privacy

Alert	Example rule
Low produce	Below bin threshold across consecutive valid observations; deduplicate until state changes.
Likely empty	Near-zero visible coverage persists at sufficient confidence; request check.
View blocked	Region cannot be assessed beyond configured period; route camera/bin check.
Camera offline	No valid frames/results beyond heartbeat interval.
Possible security event	Restricted human review cue, never a confirmed incident label.

Integrations

- Camera: RTSP/ONVIF stream and health; secure credentials and network segmentation.
- POS: store, timestamp, SKU/PLU, quantity/weight, amount, discount, transaction status. Import only needed fields.
- Inventory: product catalog, receiving, stock, restock, waste/spoilage where available.
- Notifications: dashboard first; optional approved staff channel for operational alerts.

Security controls

- Encrypt credentials and data in transit; restrict camera networks from public access.
- Use role-based permissions and stronger authentication for admin/security review.
- Log clip viewing/export, configuration changes, alert decisions, and model deployments.
- Separate aggregate analytics from restricted clips; define backup and deletion procedures.

Privacy controls

- Document a specific purpose for each camera and feature; do not repurpose footage without review.
- Prefer edge inference and event metadata over continuous cloud video upload where practical.
- Avoid face recognition, biometrics, audio capture, and persistent shopper tracking.
- Set separate retention periods for aggregate metrics and clips; restrict access and provide notices.
- Complete a local privacy/risk assessment before enabling security analytics.

Rules depend on country and context. EDPB guidance is an EU reference and ICO guidance is a UK reference; obtain local review rather than treating either as universal.

9. Roadmap and acceptance criteria

Phase	Work	Exit criteria
0. Discovery	Country, cameras, network, POS, product mapping, owners, policy.	Approved pilot scope, camera map, data access, review plan.
1. Foundation	Connect camera; health checks; zones/bins; auth and event schema.	Stable stream; documented config; raw video not retained by default.
2. Vision pilot	Traffic/fill estimates; reviewed samples; persistent alert rules.	Measured quality; staff can correct status.
3. POS/dashboard	Import POS; map products; deliver overview and detail screens.	Reconciled sales; aligned timestamps; usable reports.
4. Operations trial	Staff confirmations; measure alert quality and restock time.	Useful signals, threshold owners and workflows.
5. Security shadow	Evaluate cues without live action; restricted audit workflow.	Separate approval and acceptable false alert burden.
6. Expand	Add stores/cameras and forecasting with versioned rollout.	Per-store validation, runbook, monitoring and rollback.

Pilot measures

Category	Measure
Traffic	Count error versus manual sample; dwell error on reviewed sample.
Availability	Precision/recall of low/empty state; blocked/unknown rate; fill error.
Operations	Alert acknowledgement/restock time; duration display is low/empty.
Commercial	Sales and traffic trends; low-stock periods; promotion comparison.
Security	False alerts per camera/day, reviewer usefulness, misses in reviewed sample.

10. Team, cost drivers, decisions

Roles needed

- Store operations/product lead: workflow, thresholds, success measures, task ownership.
- Computer vision/ML engineer: calibration, labeling, evaluation, inference monitoring.
- Backend/data engineer: events, POS/inventory connectors, database, alert service.
- Frontend engineer/designer: dashboard and review workflows.
- IT/security: network, identity, access, secrets, monitoring.
- Privacy/legal and loss-prevention leads: purpose, notices, retention, review policy.

Cost drivers

- Camera count/resolution, compatibility, and whether existing views need repositioning.
- Edge hardware, hosting, video retention, network, and support.
- POS/inventory connector complexity and number of stores/SKUs.
- Annotation and staff review time to validate estimates and security cues.
- Model/API licensing, audit controls, privacy/legal review, and operational training.

Decisions needed for a build-ready specification

Decision	Why it matters
Deployment country	Applicable privacy, CCTV, employment, and evidence requirements.
Camera inventory	Coverage, stream access, resolution, hardware, blind spots.
POS/inventory vendor	Integration method, product mapping, reconciliation.
Store count and pilot site	Architecture, rollout, support, cost.
Alert recipients	Workflow, response time, escalation, permissions.
Hosting and retention	Storage/security design and operating cost.
Security feature gate	Whether it stays in shadow evaluation or limited review.

11. Assumptions, risks, references

Working assumptions

- The store has authorized access to camera streams and an approved network path.
- POS data can be exported or queried with timestamps and product identifiers.
- The first release is decision support, not autonomous inventory control or security enforcement.
- The pilot starts with one produce area and a few bins.

Risks and mitigations

Risk	Mitigation
Occlusion/poor angle	Reposition, use blocked state, dedicated views, audit conditions.
Visible fill differs from actual stock	Combine with POS/receiving/staff checks; never claim exact quantity.
POS mapping errors	Validate SKU/PLU mapping and reconcile totals first.
False alerts/alert fatigue	Persistence, confidence, deduplication, clear owner, measured quality.
Security cue misread	Human review only; no automatic action; restricted clips and policy.
Privacy/legal mismatch	Country-specific review, transparency, defined purpose, retention controls.
Display/model drift	Monitor quality; version configuration; relabel after layout changes.

References

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End of project brief